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Code: 3501/R

Faculty of Science

BCA II-Year, CBCS-III Semester Regular Examinations Nov/Dec-2024

PAPER: DATABASE DESIGN

Time: 3 Hours **Max Marks: 70**

Section-A

I. Answer any FIVE of the following questions **(5x4=20 Marks)**

1. What is an entity? Write about types of entities.
2. Explain the concept of cardinality in the context of an E-R model.
3. Write about BCNF.
4. Describe the structure of a B+ Tree.
5. List and define aggregate functions in SQL.
6. Define a PRIMARY KEY constraint and explain its purpose.
7. What is a shared lock, and when is it used in databases?
8. What is the role of a DBMS in a database environment?

Section-B

II. Answer the following questions **(5x10=50 Marks)**

9. **a)** What is database? List and explain different limitations of traditional file processing systems.

OR

b) Explain about *three schema architecture* with neat diagram.

10. **a)** Define normalization? Explain about 1NF, 2NF, 3NF and BCNF with examples.

OR

b) Explain about supertype/subtype hierarchies in detail.

11. **a)** Explain about DDL and DML commands with syntax and example.

OR

b) Explain the concept and purpose of SQL triggers.

12. **a)** What are indexing data structures? Describe the main types of indexing data structures used in databases.

OR

b) List and explain common types of file organizations used in databases

13. **a)** What is Two-Phase Locking (2PL)? Explain how it ensures serializability.

OR

b) What is a deadlock in database systems? Explain how deadlocks occur and discuss methods used to detect and resolve deadlocks.

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Code: 3502/R

Faculty of Science

BCA II-Year, CBCS-III Semester Regular Examinations Nov/Dec-2024

PAPER: OPERATING SYSTEM CONCEPTS

Time: 3 Hours **Max Marks: 70**

Section-A

I. Answer any FIVE of the following questions **(5x4=20 Marks)**

1. What are system calls? Give examples of different types of system calls.
2. List the requirements for a solution to the critical-section problem.
3. What is the purpose of swapping in memory management?
4. Briefly explain the bitmap approach to free-space management.
5. What is a Trojan horse? Explain how it poses a security risk.
6. What are the main objectives of CPU scheduling in an operating system?
7. What is the role of synchronization hardware in solving critical-section problems?
8. What is disk formatting? Why is it necessary?

Section-B

II. Answer the following questions **(5x10=50 Marks)**

9\. a) Define process? Describe about process life cycle with neat diagram.

OR

b) Consider the following set of processes, with their arrival times and burst times given below. Calculate the waiting time and turnaround time for each process using the FCFS scheduling algorithm.

Process	Arrival Time	Burst Time
P1	0	5
P2	2	3
P3	4	1
P4	6	2

10\. a) What are semaphores? Explain types and their role in solving the producer-consumer problem

OR

b) Explain the methods for handling deadlocks. Discuss deadlock prevention, avoidance, detection, and recovery in detail.

11\. a) Consider a page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3 with three frames. Using the FIFO page replacement algorithm, determine the page faults.

OR

b) Discuss contiguous memory allocation and its techniques. Explain the challenges with this allocation method and solutions like memory compaction.

12\. a) Explain the different types of access methods for files. Describe each method with examples of its usage

OR

b) Discuss file protection mechanisms. Explain how access control lists (ACLs) and user/group permissions work in modern file systems.

13\. a) Explain about system security in detail.

OR

b) Describe the security threats that can arise in programs.

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Code: 3503/R

Faculty of Science

BCA II-Year, CBCS-III Semester Regular Examinations Nov/Dec-2024

PAPER: ENVIRONMENTAL SCIENCE

Time: 3 Hours **Max Marks: 70**

Section-A

I. Answer any FIVE of the following questions **(5x4=20 Marks)**

1. Importance of Environmental studies
2. Write a note on Ecological pyramids
3. Endangered species
4. Air pollution
5. Watershed management
6. Energy flow in ecosystem
7. Thermal Pollution
8. Disaster management

Section-B

II. Answer the following questions **(5x10=50 Marks)**

9. (a) Write about use and over-utilization of surface and ground water.

(OR)

(b) Effects of modern agriculture on Environment

10. (a) Write is Ecosystem and explain its structure.

(OR)

(b) Write about Renewable and Non-renewable energy resources

11. (a) Biodiversity threats?

(OR)

(b) Write about Biodiversity conservation

12. (a) What is soil pollution? Write about its causes and control measures to reduce

(OR)

(b) Wild life protection Act?

13. (a) Ozone layer depletion?

(OR)

(b) Impact of Disasters on Environment

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Code: 3505/R

Faculty of Science

BCA II-Year, CBCS-III Semester Regular Examinations Nov/Dec-2024

PAPER: JAVA PROGRAMMING

Time: 3 Hours **Max Marks: 70**

Section-A

I. Answer any FIVE of the following questions **(5x4=20 Marks)**

1. Difference between C & JAVA?
2. What is an Event?
3. What is a Package?
4. What are the different Data types in Java?
5. What is Vector?
6. What is Swing?
7. What is Dead Lock?
8. What is Stream?

Section-B

II. Answer the following questions **(5x10=50 Marks)**

9. (a) Explain Java Features?

OR

(b) Explain method overloading & constructor overloading?

10. (a) What is an Array? Explain about java.util.array class is used to create array using program?

OR

(b) Explain method overriding with program?

11. (a) What is Exception. Explain how to handle Exceptions?

OR

(b) Explain Thread and its life cycle?

12. (a) Explain GUI(AWT) controls with help of program?

OR

(b) Explain Applet with program?

13. (a) Explain Reader & writer class with help of program?

OR

(b) Explain Collection frame work with program?

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Code: 3505/R

Faculty of Science

BCA II-Year, CBCS-III Semester Regular Examinations Nov/Dec-2024

1 ㄩ/ **PAPER: APPLIED MATHEMATICS**

Time: 3 Hours **Max Marks: 70**

Section-A

I. Answer any FIVE of the following questions (5x4=20 Marks)

1. Define the Eular`s theorem and verify $f(x,y) = ax^2 + 2hxy + by^2$
2. If $u = x^3+y^3$ where $x= a \cos t$, $y= b \sin t$ then evaluate $\frac{\partial u}{\partial t}$ and verify the result.
3. Define the consistent , Inconsistent equations.
4. Define Linear Dependent , Linear Independent Vectors.
5. Find the value D^5 of $D =\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$
6. Show that the intersection of any two subspace of a vector space V is subspace of V.
7. Define Eigen value, Eigen vector of a matrix A. find Eigen value of $matric \begin{pmatrix} 2024 & 0 \\ 0 & 2025 \end{pmatrix}$
8. If $u = y\, e^{xy} \sin x$ then find $\frac{dy}{dx}$, at O(0,0)

Section-B

II. Answer the following questions (5x10=50 Marks)

9. (a) If $u= \log (x^3+y^3+z^3-3xyz)$ then show that $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}=\frac{-3}{(x+y+z)2}$

OR

(b) State and prove Euler's theorem on homogeneous functions.

10. (a) Prove that if the perimeter of a triangle is constant, its area is Maximum when the triangle is equilateral

OR

(b) If $ax^2+2hxy+by^2+2gx+2fy+c=0$ then prove that $\frac{d2y}{dx2}=\frac{abc+2fgh-af2-bg2-ch2}{(hx+by+f)2}$

11. (a) Check the given linear equation consistence or Inconsistent $x-3y+2z+2w =1;2x-2z =3$; $4x-6y+2z+4w =6$.

OR

(b) Determine whether the vectors $V_1=\{-1,0,-2\},V_2=\{2,3,-4\},V_3=\{-3,-5,1\}$Linear Independent or Linear Dependent. In R^3.

12. (a) $A =\begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix}$ then find Eigen values and Eigen Vectors

OR

(b)Find the Characteristic equation , and values of the matrix $\begin{pmatrix} 1 & 0 & -1 \\ -2 & 3 & +1 \\ 0 & 6 & 0 \end{pmatrix}$

13. (a) Diagonalize the matrix $\begin{pmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ -1 & -2 & 2 \end{pmatrix}$; if possible.

OR

(b) An n x n matrix A is diagonalizable if and only if it has n linearly independent Eigen Vectors.

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